Compute Market Playbook

A Framework for Standardized AI
Compute Pricing and Derivatives

A patent-pending methodology creating a universal unit of measurement for AI accelerator performance — enabling the first standardized derivatives market for compute.

YGGDRASIL FINANCIAL TECHNOLOGIES

Seed Round · $3M Post-Money SAFE · $15M Cap

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EXECUTIVE SUMMARY

With hyperscaler capital expenditure reaching $365 billion in 2025[1] and projected to exceed $600 billion by 2026,[2] AI compute has emerged as the most strategically important resource of the 21st century — yet enterprises lack standardized price discovery, hedging instruments, and efficient resource allocation tools.

This playbook presents a framework for establishing standardized AI compute pricing infrastructure — anchored by a patent-pending normalization methodology, a working live data platform, and a live dataset spanning 28 cloud providers and 56 accelerators.

KEY FACTS

1. **Market Demand Exists:** Global public cloud end-user spending reached $723 billion in 2025,[3] with GPU rental price swings of up to 88%[4] and no hedging mechanisms available. In January 2026, AWS raised GPU instance prices 15% overnight with no advance notice[5] — enterprises had no recourse.
2. **Historical Precedents Validate the Approach:** WTI crude oil, VIX futures, electricity markets, and carbon trading all became multi-billion dollar markets after transparency mechanisms were established. Southwest Airlines saved $3.5B hedging fuel costs.[6] The pattern is consistent and repeatable.
3. **The Platform Is Built:** 5.6M+ price ticks across 28 cloud providers, 56 accelerators scored, 34 live API endpoints, GPU benchmark suite, and a patent-pending normalization methodology with Spearman $\rho > 0.96$ against MLPerf benchmarks. This is not a pitch for an idea.
4. **Institutional Interest Is Real:** Goldman Sachs Futures Sales Trading (MD + VP level) has engaged in initial conversations on compute derivatives. $14.2B in GPU-backed debt (CoreWeave)[7] proves compute is already institutional collateral.
5. **Competitive Validation:** Three competitors have raised $45M+ combined (Ornn $5.7M, Silicon Data $4.7M, Architect $35M). None have cross-chip normalization. None have IP protection on the methodology layer.

The Ask: $3M post-money SAFE ($15M cap). Hire 3 engineers + fractional CCO from Day 1. Deploy systematic benchmarking. Launch enterprise API sales. Publish on Bloomberg/Refinitiv. 26 months runway at zero revenue. $100K MRR by Month 20–24 on data subscriptions alone. Derivatives settlement is pure upside.

THE PROBLEM: WHY COMPUTE NEEDS A STANDARD

Market Opacity

Global AI spending hit $1.5 trillion in 2025 and will reach $2.5 trillion in 2026 (Gartner).[8] GPU compute is the underlying commodity — yet it operates with the same pricing opacity that characterized oil markets before WTI standardization:

The AI compute market exhibits characteristics that typically drive derivative market formation:

Metric	Data Point	Implication
H100 Spot Price Volatility	–88% after launch	Extreme price swings create hedging demand
Cloud GPU Cost Dispersion	46% coefficient of variation	Same GPU, wildly different prices across providers
Enterprise GenAI Spending	$37B (up from $11.5B in 2024)	Massive exposure without hedging tools
GPU-Backed Debt Outstanding	$14.2B (CoreWeave alone)	Institutional collateral with no reference price
AWS Price Hike (Jan 2026)	15% overnight, no notice	Zero hedging infrastructure for enterprises

An NVIDIA H100 and an AMD MI300X are fundamentally producing the same commodity — compute — but through different architectures (CUDA vs ROCm) and different methods of delivering that compute. An H100 hour at one provider costs $2.15; at another, $4.20.[9] An MI300X delivers more raw FLOPS but with different software overhead. Yet there is no common denominator to compare them. Without a standard unit of measurement: no futures, no swaps, no hedging.

The WTI Precedent — Oil Before 1983

Before WTI standardization, the oil market was fragmented across incomparable crude grades — West Texas Sour, Brent, Dubai, Arab Light — each priced regionally with no common reference. Airlines hedged jet fuel using crude oil forward contracts, but with only ~0.63 correlation — expensive, illiquid, and riddled with basis risk.[10]

WTI defined a **standard grade** based on physical properties (API gravity, sulfur content) — and that grade became the contract denomination for the American oil derivatives market. Today WTI underpins 2M+ contracts traded daily with $180B+ in daily notional value.

Standardization made hedging practical and scalable. Southwest Airlines saved $3.5B using WTI-referenced futures to hedge fuel costs — while airlines that didn't hedge faced the 2008 fuel crisis unprotected, collectively losing $5.8B as five carriers filed for bankruptcy.

GPU compute today = oil in 1982. CU is the standard.

Historical Precedents — Every Commodity Follows This Path

Market	Standardization	Derivatives	Scale
Oil (WTI)	1983 — API gravity	NYMEX futures 1983	$65T+ annual notional
Electricity	PJM hub 1997	Nodal futures/ swaps	$500B+ annual notional
Volatility	VIX formula 1993	Futures 2004	$1.5T+ annual notional
Carbon	EU ETS 2005	Futures + options	$949B annual notional
Bitcoin	CME ref rate 2016	Futures 2017	$15T+ annual notional
Compute	**CU — 2026**	**OTC → CME futures**	**$50–100B+ at maturity**

Sources: LSEG Carbon Market Year in Review (Feb 2024); CNBC/CME Group Bitcoin futures launch (Dec 2017); CBOE VIX history; FERC Order No. 888 (1996) for electricity deregulation; Nord Pool Trading Figures 2023.

Every one of these markets followed the same sequence: **standard → index → OTC → exchange-listed derivatives.** The VIX illustrates the trajectory most clearly: from a formula published in 1993,[11] to futures in 2004,[12] to options in 2006, to 53.7 million contracts per year[13] — a 583× increase from launch. CBOE built a $17.8 billion company on that single index.

The standardization layer is always the most defensible position. It is embedded in every contract definition and referenced by every downstream product.

The CoreWeave Problem — Depreciation Without Data

CoreWeave is claiming **6-year useful life depreciation** on GPU assets. Industry consensus is approximately 2 years. No one can verify either claim because **no standardized performance degradation data exists.**

- $14.2 billion in GPU-backed debt (Blackstone, PIMCO, BlackRock, Morgan Stanley)[14]
- $1.1 billion AAA-rated GPU asset-backed security — spread inside 110 bps
- 250,000+ NVIDIA GPUs financed as institutional collateral
- Lenders pricing risk with internal models — no external reference rate

GPU performance degrades over time — thermal stress, HBM reliability decay, clock throttling from aging power delivery. A GPU's Day 1 performance ≠ Day 1,000 performance. Without systematic measurement, lenders cannot price collateral aging risk.

Yggdrasil solves this directly. By re-running physical benchmarks on the same hardware monthly, the platform produces **degradation curves** — the performance decline trajectories that lenders need to price GPU-backed debt accurately. No other company systematically measures hardware aging.

Who Needs This

Segment	Use Case
Cloud Providers	Transparent pricing builds trust. Performance benchmarking validates fleet quality against competitors.
Enterprises	Budget forecasting. Vendor comparison on a performance-adjusted basis. Cost optimization across architectures.
GPU Lenders	Collateral valuation for $14.2B+ in GPU-backed debt. Depreciation curve data. Insurance pricing.
Trading Firms / Investment Banks	Hedging GPU exposure. Building derivatives positions. Futures curve construction from degradation data. Structured product creation.
Hardware Makers	Lifecycle planning. Competitive positioning. Fleet management intelligence.

THE SOLUTION: THE COMPUTE UNIT

Defining the Compute Unit

The **Compute Unit (CU)** is a patent-pending normalization methodology that assigns every AI accelerator a deterministic score derived from physical hardware specifications. It is the **grade specification** for compute — the equivalent of WTI's API gravity standard for crude oil.

> **An auditor can reproduce a CU score from a spec sheet. They cannot reproduce a benchmark run.** This distinction is the entire basis for settlement-grade derivatives.

CU is computed from four hardware specification inputs, each publicly available from manufacturer datasheets:

Input	Description
FP Throughput	Multi-precision composite (FP16, FP8, FP32) weighted by workload relevance. Training emphasizes compute intensity; inference emphasizes throughput.
Memory Bandwidth	HBM or SRAM bandwidth — the primary bottleneck for inference workloads. Bandwidth determines how fast model weights can be served.
VRAM Capacity	Total accelerator memory — determines maximum model size that can be hosted without sharding across multiple chips.
Interconnect BW	NVLink, ICI, or custom fabric — determines multi-chip scaling efficiency for distributed training and tensor parallelism.

These inputs are combined using a **calibrated geometric mean model** with physics-constrained exponents. The exponents are optimized via differential evolution against MLPerf training benchmarks, vLLM inference benchmarks, and TensorRT-LLM measurements, achieving Spearman rank correlation $\rho > 0.96$.

Three output scores:

- **CU** — unified compute score (harmonic mean of TCU and ICU)
- **TCU** — training compute unit (compute-intensity weighted: $\alpha > \beta$)
- **ICU** — inference compute unit (memory-bandwidth weighted: $\beta' + \gamma' > \alpha'$)

The physics constraints ensure that TCU always weighs raw compute more heavily (training is compute-bound), while ICU always weighs memory bandwidth more heavily (inference is memory-bound). These constraints are structural — they cannot be violated by any calibration.

Published as a **daily reference rate + real-time index.** Every score is deterministic, auditable, and reproducible by any third party with access to the manufacturer's spec sheet.

Why Deterministic Matters for Derivatives Settlement

Requirement	Token Benchmarks	Price Index	Spec-Based CU
Deterministic	No: varies by test	Yes: observable	**Yes: fixed formula**
Auditable by regulators	Hard: reproduce env	Yes: price feeds	**Yes: published specs**
Cross-GPU comparable	Limited	No: separate indices	**Yes: unified score**
Training + Inference	No (inference only)	No	**Yes (TCU + ICU)**
Available pre-launch	No: need hardware	No: need trading	**Yes: from spec sheet**
IOSCO-compliant	Governance challenges	Possible	**Designed for it**
Version-stable	No: software changes	No: index updates	**Yes: formula is fixed, auditable**

The Six-Layer Market Architecture

A complete compute commodity market requires six layers — from raw hardware specifications to exchange-traded derivatives. Yggdrasil owns the foundation layers that every subsequent layer depends on.

MARKET ARCHITECTURE

Layer 1	Hardware Specifications	*Manufacturer datasheets*
Layer 2	CU Standardization	***Yggdrasil (patent-pending)***
Layer 3	Benchmark Validation + Degradation	***bm-suite physical measurement***
Layer 4	Price Discovery	***28-provider collection pipeline***
Layer 5	Index Construction	***CU-normalized reference rates***
Layer 6	Derivatives	*OTC swaps → CME-listed futures*

Yggdrasil owns Layers 2–5 — the entire value chain from raw specifications to tradeable index. This is the infrastructure that derivatives settle against.

The degradation layer (Layer 3) is a key differentiator. By re-running physical benchmarks on the same hardware monthly, Yggdrasil tracks performance decay — producing **eCU** (empirical Compute Unit) scores that diverge from spec-based CU over time. The **CU–eCU spread** quantifies the gap between theoretical maximum and actual delivered compute:

- **Provider-specific** — same GPU model, different eCU at AWS vs Lambda vs VAST
- **Time-varying** — degradation shows up as eCU declining while CU stays constant
- **Tradeable** — institutional-grade data for futures curve construction and GPU-backed loan pricing

Multi-Architecture Coverage

CU scores every accelerator architecture — GPU, TPU, LPU, WSE, hardwired ASIC — on a single unified scale:

Hardware	Class	Method	TCU	ICU	CU
8× H100 SXM	GPU	Reference	1.000	1.000	1.000
8× MI300X OAM	GPU	Published Spec	2.172	2.821	2.454
TPU v5p Superpod	TPU	Published Spec	5,202	94,480	9,860
Groq GroqRack	LPU	Published Spec	N/A	2.07	2.07
Cerebras WSE-3	WSE	Tok/s Bridge	0.37	varies	varies

All scores relative to 8× H100 SXM DGX cluster (CU = 1.000). The methodology handles exotic architectures (Groq's SRAM-only LPU, Cerebras's wafer-scale engine) via scoring method variants documented in the patent application.

Financial Contract Underlyings

The CU framework supports a tiered underlying structure for derivatives:

Contract Type	Underlying
GPU-specific hedging	Raw $/hour per GPU model — direct price hedging for enterprises renting specific hardware (e.g., H100 futures, MI300X forwards)
Cross-architecture contracts	CU, TCU, and ICU scores — normalized indices enabling hedging across GPU architectures on a single scale
Broad market indices	CU/TCU/ICU weighted by eCU, eTCU, and eICU (empirical scores from benchmarks) — accurately representing real deliverable compute per accelerator, not just theoretical specs

The weighting methodology between spec-based scores (CU) and empirical measurements (eCU) is proprietary. This mirrors how S&P weights constituents in the S&P 500 — the methodology is the moat.

INTELLECTUAL PROPERTY

Patent Status

Item	Status
Provisional Patent	Filed January 9, 2026 — priority date secured
Non-Provisional	Filed. USPTO review in progress. Track One prioritized examination recommended (9-month timeline).
Claims Cover	Cross-chip normalization formula; CU/TCU/ICU decomposition; multi-architecture scoring (GPU, TPU, LPU, WSE, HWASIC); physics-constrained exponents; degradation curves; virtual cluster bridge; physical reference linkage between hardware specs and scoring methodology

Freedom to Operate

No competing patents exist on cross-chip normalization methodology (USPTO search conducted). Competitors operate in adjacent but non-overlapping spaces:

- **Ornn:** Price aggregation — no normalization IP, no patent filed
- **Silicon Data:** Empirical benchmarking (SiliconMark) — measures instance performance, no normalization methodology, no patent
- **SemiAnalysis / InferenceX:** Open-source benchmark — no IP claims

IP as Moat

The grade specification is the most defensible layer of commodity market infrastructure. Once adopted:

- **Embedded in every contract definition** — switching costs are enormous
- **Referenced by every downstream product** — every index, ETF, swap, and future
- **Only layer requiring novel IP** — price aggregation is obvious; cross-chip normalization from hardware specifications is not
- **Patent priority date locks out competitors** — any future filing is subordinate to the secured priority date

WHAT'S BUILT TODAY

This is not a pitch for an idea. Everything below was solo-built by the founder in under 6 months. $3M takes this from MVP to enterprise-ready to revenue-generating.

Component	Status	Detail
bm-suite	Live	Benchmark suite measuring 5 performance dimensions (GEMM FLOPS, memory bandwidth, VRAM capacity, interconnect throughput, inference tok/s). NVIDIA + AMD support. Docker-based with thermal gating, parallel GPU runner, and calibration scripts.
Collection	Live	Price collection pipeline across 28 cloud providers. 5.6M+ price ticks ingested. Base collector framework with retry logic, change detection, and forward-fill.
CU Engine	Live	Deterministic scoring engine. v3 physics-constrained weights. 56 accelerators scored. Calibrated against MLPerf and vLLM benchmarks. Spearman $\rho > 0.96$.
API	Live	FastAPI with 34 endpoints. Zero-copy BookStore serving architecture. API key authentication. Async request handling.
Database	Live	TimescaleDB with 5.6M+ ticks. 100-field schema. 7 continuous aggregates. 9 optimized indexes. 77 materialized data books.
Dashboard	Live	React 19 + Vite + ECharts + Tailwind CSS. 7 pages with command bar, filter bar, and real-time data display.
GPU Specs	Live	gpu_specs.json covering all datacenter-grade accelerators and top-tier consumer-grade. v2.2.0. Cross-referenced against manufacturer datasheets.

BUSINESS MODEL

Revenue Streams

Stream 1 — Data API Subscriptions

Tier	Price	Access
Free	$0/mo	Index-level CU/TCU/ICU headline numbers. 1-hour delay. 100 API calls/day. Signup required for lead generation.
Pro	$3K/mo	Real-time REST API. Per-GPU and per-provider breakdowns. Full historical archive. 10K calls/day. CSV/JSON export.
Enterprise	$10K/mo	WebSocket real-time streaming. Per-instance granularity. eCU + degradation data. Raw benchmark results. Settlement-grade data with audit trail.

Pricing validated against: Nasdaq Data Link $3K/mo, FactSet API $2–6K/mo, Platts $5K/mo. Silicon Data's H100 rental index is already distributed on Bloomberg ALTD,[15] validating the terminal distribution pathway.

Stream 2 — Terminal Distribution ($500K–2M/yr potential)

Bloomberg ALTD and Refinitiv distribution via dxFeed. Silicon Data validated this pathway — their H100 rental price index is already on Bloomberg terminals. Revenue model: flat annual fee plus usage-based licensing.

Stream 3 — Derivatives Settlement (long-term upside)

Product	Rate	Precedent
OTC Swaps	30–40 bps notional	SOFR/LIBOR reference licensing model
Exchange Futures	$0.10/contract	CME standard per-contract royalty
AUM Products (ETF/ETN)	3–5 bps of NAV	S&P receives 3 bps from SPY ($120M/yr)

TAM / SAM / SOM — By Revenue Stream

Each revenue stream has its own market sizing. Streams are additive and sequential — data subscriptions fund operations, terminal distribution builds credibility, OTC swaps are the inflection point, exchange listing is the end-state.

Stream 1 — Data Sales (API Subscriptions)

Metric	Value	Derivation
TAM	$60M/yr	Bottom-up by segment: 500 GPU cloud providers × $3K/mo = $18M; 150 cloud brokers/MSPs × $3K/mo = $5.4M; 300 quant desks × $10K/mo = $36M. Pricing anchored below comparables: Orbital Insight satellite data = $200–300K/yr per fund; FactSet enterprise seat = $25–50K/yr.
SAM	$25–42M/yr	500 active providers/brokers + 200 quant desks — reachable in 3–5 years with Bloomberg distribution
SOM Yr 1	$120–300K	5–8 provider subs ($3K/mo), 0–1 quant desks. Grows to $540K–1.2M in Year 2 with Bloomberg.

Stream 2 — Terminal Distribution (Bloomberg ALTD / Refinitiv)

Metric	Value	Derivation
TAM	$150–200M/yr	Alternative data market on financial terminals ($2B total alt data, 8–10% compute-relevant). Nasdaq Link institutional data: $100–500K/yr per subscriber.
SAM	$3–5M/yr	Annual licensing for novel commodity index on ALTD — comparable datasets: $1–5M/yr flat + usage
SOM Yr 1	$0	12–18 month onboarding cycle. Year 2: $500K–1.5M

Stream 3 — OTC Bilateral Swaps

Precedent: VIX index licensing generates $40M+/yr for CBOE. WTI crude — ~35% of physical production hedged at maturity. Energy bilateral OTC references Henry Hub/WTI at 2–5 bps. GPU cloud TAM: $723B (Gartner 2025), growing 25–40%/yr. The table below uses *projected* TAM at each milestone year, mapped to the oil market's hedging adoption trajectory.

Year	% Hedged	Notional	Fee	Revenue	Comparator
Y3	0.05%	$1B	35 bps	$3.5M	First bilateral swaps
Y5	0.5%	$17B	25 bps	$42M	VIX-scale licensing
Y7	3%	$164B	15 bps	$245M	Exchange listing begins
Y10	15%	$1.5T	10 bps	$1.5B	Energy OTC analog
Y15	**35%**	**$6.2T**	**6 bps**	**$3.7B+**	**Oil-equivalent maturity**

Stream 4 — Exchange-Listed Futures

Metric	Value	Derivation
TAM	$50–200M/yr	CME earns $0.05–0.25/contract. VIX futures royalties: $15M/yr for CBOE. Bitcoin futures: 9 years OTC → CME. Electricity: 5–7 years.
SAM	Same as TAM	Year 4–6 milestone event, not a revenue ramp
SOM	Not Yr 1–3	OTC volume must validate demand first. Exchange listing is the end-state, not the plan.

Combined	TAM	SAM	SOM Yr 1
Data Sales	~$60M	$25–42M	$120–300K
Terminal	$150–200M	$3–5M	$0 (Yr 2: $500K+)
OTC Swaps	$100–375M	$10–28M	$0 (Yr 3: ~$3.5M)
Exchange Listing	$50–200M	same as TAM	N/A — Yr 4–6

The OTC and derivatives TAM is not speculative. VIX-related products generate $40M+/yr in licensing revenue for CBOE.[16] S&P earns $500M+/yr from index licensing. Compute is a larger underlying market than volatility.

Year 1 SOM is entirely data subscriptions — terminal and OTC revenue kicks in Year 2–3 and eventually dwarfs the data business.

CU vs eCU — The Premium Data Product

The **CU–eCU spread** is the premium data product that no competitor can replicate:

- **CU** = theoretical maximum silicon capability (deterministic, from spec sheets)
- **eCU (empirical Compute Unit)** = actual measured performance from bm-suite Docker benchmarks run on real rented hardware
- **CU–eCU Spread** = quantified gap between theoretical spec and real-world delivery — the "software tax" and degradation signal

An AMD MI300X might show CU of 2.45 but eCU of 1.7 because ROCm software overhead eats 30% of theoretical performance. An NVIDIA H100 SXM might show eCU of 0.95 because CUDA is mature. This spread is provider-specific (same GPU, different eCU at AWS vs Lambda vs VAST), time-varying (degradation shows up as eCU declining while CU stays constant), and tradeable.

COMPETITIVE LANDSCAPE

Competitor Overview

Three companies have raised $45M+ combined in the compute data/pricing space. None have cross-chip normalization. None have IP protection on the methodology layer.

Company	Raised	Approach	Gap
Ornn	$5.7M	Price aggregation — HCPI for H100	No cross-chip normalization. H100 index and B200 index are unrelated numbers. Cannot build cross-GPU structured products.
Silicon Data	$4.7M	Empirical benchmarking (SiliconMark)	Benchmarks only — no normalization. Non-deterministic (same GPU scores differently each run). Inference-only. Cannot be audited from spec sheet.
Architect	$35M	Exchange platform for compute	Exchange infrastructure only — uses Ornn's index as settlement benchmark. No proprietary methodology. Dependent on upstream index quality.

The oil parallel: Before WTI standardization, oil companies published proprietary quality metrics. Once API gravity became the standard grade spec, every proprietary metric became subordinate. CU is the API gravity of compute — the competitors above are building on sand until a normalization standard is adopted.

Silicon Data Deep Dive — SiliconMark

Silicon Data's SiliconMark is an empirical benchmarking tool that measures FLOPS, memory bandwidth, and inference latency on rented GPU instances. It is backed by DRW and Jump Trading — two of the most sophisticated quantitative trading firms — and their H100 rental price index (SDH100RT) is already on Bloomberg ALTD.[17]

The key distinction: SiliconMark measures **what a specific GPU instance delivers on a specific day.** CU measures **what a hardware architecture is capable of delivering based on its physical specifications.** These are fundamentally different products:

Dimension	SiliconMark (SD)	CU (spec-based)	eCU (empirical)
Method	Run workload, measure output	Calculate from spec sheet	Run bm-suite Docker benchmarks on rented hardware
Determinism	Non-deterministic (varies by run)	Fully deterministic (permanent)	Non-deterministic (captures real-world variance)
Coverage	Inference only	Training (TCU) + Inference (ICU)	Training + Inference + degradation
Pre-launch	No — needs hardware	Yes — from spec sheet	No — needs hardware
Audit	Reproduce test env	Reproduce from public specs	Reproduce via open Docker benchmarks
Settlement	IOSCO challenges	Designed for IOSCO	Calibration input to CU

SiliconMark is quality assurance. CU is the grade specification. Both are needed. SD's empirical data feeds *into* CU calibration. This is a complementary relationship, not a competitive one.

Note: SiliconMark's benchmark approach is functionally equivalent to Yggdrasil's eCU — empirically-run workloads producing hardware-specific performance scores. The distinction: eCU feeds into the CU normalization standard and produces a cross-architecture index; SiliconMark does not. Covered in detail on the next page.

The Complementarity Stack

Every mature commodity market has three infrastructure layers:

1. **Grade Specification** (CU / Yggdrasil) — defines the contract denomination
2. **Spot Pricing** (Ornn today; Yggdrasil's GPU Price Index) — tracks what the market pays for each grade
3. **Quality Assurance** (SiliconMark / SD today; Yggdrasil's eCU benchmarks) — verifies instances perform to grade

Oil has API gravity specs, Platts/Argus pricing, and assay testing. Compute will need all three — and **Yggdrasil covers all three layers.** The CU scoring engine provides grade specification, the GPU Price Index delivers real-time spot pricing, and eCU empirical benchmarks serve as quality assurance. **The strategic question is: who owns the grade specification?** That is the most defensible position because it is embedded in every contract definition and is the hardest to displace once adopted.

Once CU becomes the CME settlement benchmark, competitors have no choice but to integrate with Yggdrasil to participate in compute-based financial derivatives. The standard owner absorbs the ecosystem — pricing feeds must denominate in CU, quality tests must calibrate against CU grades, and every contract settles on the Yggdrasil index. This is the S&P playbook: own the standard, and the ecosystem builds around you.

Why No One Else Can Do This

- **Architect's $35M exchange needs a settlement index.** They currently use Ornn's HCPI — which has no cross-chip normalization. CU is the upgrade they need but cannot build (patent priority blocks them).
- **Neither Ornn nor SD have IP protection** on normalization methodology. Their moats are data and relationships, not technology.
- **NVIDIA won't publish their own standard** — standardizing performance across architectures would validate AMD/TPU competitors. Antitrust risk is obvious.
- **Patent priority date locks this door.** Any competitor filing a cross-chip normalization patent after the secured priority date is subordinate.

GO-TO-MARKET STRATEGY

Phase 1: Data Customers (Month 1--6)

- **Target:** Neoclouds, GPU brokers, cloud comparison platforms, quant firms
- **Named leads:** CoreWeave, Lambda Labs, Together AI, Vultr, Shadeform
- **Sales motion:** Founder direct outreach + personal network connections. "Pricing transparency" pitch.
- **Conversion funnel:** Free tier sign-up → data exploration → Pro upgrade
- **Quant firm funnel:** Find alpha signals in CU Index data (NVDA earnings correlation, cloud margin divergence, inference vs training spread) → publish research paper on SSRN → quant desks discover it → validates $10K/mo subscription value. Same model as Orbital Insight: satellite imagery → oil tanker alpha → fund subscriptions.
- **Goal:** 5–8 Pro + 1–2 Enterprise customers. $25–44K MRR by Month 6.

Phase 2: Terminal Distribution (Month 9--15)

- Bloomberg ALTD submission — Silicon Data validated this exact pathway
- Refinitiv distribution via dxFeed
- Revenue: annual licensing fee + per-query usage
- **Credibility inflection:** CU tickers on Bloomberg terminals. Every bank trading desk with a terminal discovers CU organically. This drives Phase 3.

Phase 3: Institutional / Derivatives (Month 9--18)

- **Goldman Sachs:** Two exploratory meetings with Futures Sales Trading (MD + VP level). Direction received: build long and short side from public prices. Product built to that specification — independently, not as a contracted engagement.
- **OTC path:** Banks get free CU index data → create bilateral OTC swaps referencing CU → pay licensing fee (30–40 bps notional + minimum annual). This is exactly how SOFR, LIBOR, and VIX work.
- **CME engagement:** Once OTC swap volume validates demand.
- **Path:** OTC swaps → prove demand → CME/CBOE listing.

Phase 4: Exchange Listing + Expansion (Month 18+)

- **CME/CBOE listing:** Once OTC swap volume validates demand, pursue exchange-listed compute futures
- **Overseas expansion:** Asian compute markets (China, Japan, Korea) — trillions in spend with zero hedging infrastructure
- **Adjacent infrastructure:** Extend CU methodology to RAM, storage, and networking — every infrastructure component that can be commoditized
- **Sector-specific indices:** HPC Index, Rendering Index, Genomics Index — each a separate tradeable benchmark with its own derivatives market

Customer Pipeline — Month 6 Target: 10 Paid Subscribers

At seed-stage B2B with cold outreach, expect 25–35% pipeline-to-close. To hit 10 customers by Month 6, we need 30+ qualified leads in the funnel by Month 2.

Tier 1: GPU Brokers & Comparison Platforms (Easiest Sell)

These companies resell or compare compute — CU-normalized pricing is immediate competitive advantage.

#	Company	Why They Buy	Close %
1	Shadeform	GPU broker, aggregates 20 providers. CU scores show customers *value* not just *price*. Already scrapes pricing — $3K/mo for clean + normalized data is a no-brainer.	60%
2	FluidStack	Decentralized GPU marketplace. Heterogeneous supply (consumer + datacenter). CU normalizes their entire catalog.	50%
3	RunPod	GPU cloud marketplace. Competes on price transparency. CU data embedded in comparison UI is a marketable feature.	45%
4	Vast.ai	Spot GPU marketplace. Entire value prop is finding cheap compute — CU price-per-unit shows “best value” not just “cheapest.”	40%
5	Infracost	Cloud cost management. Already sells to enterprises. GPU pricing data is a natural product extension.	35%

6	Antimetal	Cloud cost optimization, automated reserved instance management. GPU CU data extends their optimization beyond standard cloud.	30%

Tier 2: Neoclouds (Competitive Intelligence)

These companies sell compute — they want to know how they are priced relative to 27 other providers.

#	Company	Why They Buy	Close %
7	Lambda Labs	GPU cloud, $400M revenue. Competes against AWS/GCP on GPU specifically. Needs pricing intel on competitors to set rates.	45%
8	Together AI	Inference-focused GPU cloud. Needs spot price movements across competitors for dynamic pricing strategy.	35%
9	Vultr	Multi-architecture provider (MI300X + MI325X + NVIDIA). CU cross-GPU normalization directly applicable to their pricing.	35%
10	CoreWeave	$15B RPO, largest neocloud. Pricing team needs market data for contract rates. Big enough to build internally — harder close.	25%
11	Crusoe Energy	GPU cloud on stranded energy. Unusual cost structure — needs market pricing to set competitive rates.	30%
12	Voltage Park	Leases GPU clusters. CU-normalized comps help price multi-year lease contracts fairly.	30%

Tier 3: Quant / Financial (Higher ACV, Longer Cycle)

$10K/mo tier. Takes longer but validates the premium product and the SSRN paper funnel.

#	Company	Why They Buy	Close %
13	Two Sigma	Quantitative fund, massive GPU spend. Buys for (a) trading signals in NVDA/AMD models and (b) own GPU procurement optimization.	15%

14	Citadel Securities	Market maker, huge infra spend. GPU pricing data as alt-data signal for semiconductor and cloud stock factor models.	15%
15	Point72	Dedicated data acquisition team evaluates alt-data providers constantly. GPU fleet capacity is a novel signal.	15%

Tier 4: Already on Our Index (Built-In Leverage)

These providers are already tracked on the CU Index — we publish their pricing. They have inherent motivation to subscribe: see how they rank, monitor competitors, and get the full normalized dataset. The pitch is: "We are already showing your pricing to the market. Subscribe and you get the full picture — every provider, CU-normalized, real-time."

#	Company	Why They Buy	Close %
16	Hot Aisle	AMD MI300X provider ($1.99/hr). Needs cross-architecture CU comparison to market against NVIDIA-only competitors.	40%
17	TensorWave	AMD MI300X/MI325X/MI355X. Only multi-gen AMD provider — CU normalization shows their value vs NVIDIA.	35%
18	Denvr Dataworks	GPU cloud. Already on our index — subscribe to see full competitive positioning across 28 providers.	30%
19	Massed Compute	GPU marketplace. CU data helps them price heterogeneous supply.	30%
20	Latitude.sh	Bare-metal GPU cloud. Needs market pricing intel for their GPU offerings.	25%
21	Scaleway	European GPU cloud. Needs US + EU market data for transatlantic pricing strategy.	25%
22	Excess Supply	GPU capacity marketplace. CU normalization directly applicable to their cross-provider listing model.	30%
23	Cudo Compute	Decentralized GPU cloud. Heterogeneous supply — CU scores help them price fairly across hardware.	30%
24	IMWT	GPU cloud. Competitive intel — already being compared on our index.	20%

25	Verda	GPU cloud. Same competitive intel play.	20%
26	Paperspace (DigitalOcean)	GPU cloud under DO umbrella. Already on our index. Enterprise pricing team needs market intelligence.	25%

Tier 5: Expansion Pipeline (Additional Targets)

Additional leads outside our tracked provider list.

#	Company	Why They Buy	Close %
27	TensorDock	Budget GPU marketplace. CU value-scoring differentiates them from pure-price competitors.	35%
28	Datacrunch	European GPU cloud. Needs US market pricing for transatlantic customers.	25%
29	Akash Network	Decentralized compute marketplace. Heterogeneous supply — CU normalization solves their core pricing problem.	30%
30	Modal	Serverless GPU compute. Needs pricing data for rate-setting against on-demand providers.	25%
31	Replicate	Inference-as-a-service. Buys GPU capacity from providers — CU data informs procurement.	25%
32	Salad.com	Distributed GPU, consumer hardware. CU normalization makes their heterogeneous fleet comparable.	25%
33	Genesis Cloud	European GPU cloud. Competitive intel for cross-market pricing.	20%

Conversion Math

Tier	Leads	Avg Close	Expected Closes
Brokers / Platforms	6	43%	2–3
Neoclouds	6	33%	2
Quant / Financial	3	15%	0–1
Already on Index	10	29%	2–3
Expansion Pipeline	8	26%	2
Total	**33**	**30%**	**8–11**

8–11 from outbound pipeline + 2–3 from inbound (referrals, conference leads, content marketing) = **10–14 paying customers by Month 6.** The "already on our index" tier is the strongest leverage — we are publishing their data regardless, and the pitch converts naturally from "we track you" to "subscribe and see everyone else."

Quant Firm Conversion Funnel

Separate from direct sales. Runs in parallel during IOSCO compliance wait:

1. Find alpha signals in CU Index data — NVDA earnings correlation, cloud margin divergence, inference vs training spread, provider capacity signals
2. Write research paper with factor model results
3. Publish on SSRN — quant desks discover it organically
4. Actively pitch quant funds by pointing to the published paper — warm outreach, not cold sales
5. Inbound from paper → demo → $10K/mo subscription validates premium tier
6. Same model as Orbital Insight: satellite imagery → oil tanker alpha → $200–300K/yr fund subscriptions

FINANCIAL MODEL

Use of Funds — $3M

Category	Alloc.	Detail
One-Time Startup Costs	5.3%	$158K — CFTC opinion $35K, E&O/D&O insurance $15K, 3 workstations $15K, on-prem server $12K, office buildout $18K, Vanta SOC 2 setup $15K, patent $1.5K, legal/IP $8K, corporate setup $3.5K, hiring $10K, buffer $25K.
Year 1 Operations	36.7%	$1,102K — Salaries $682K (CEO $140K, Eng1 $150K, Eng2 $150K, Eng3 $140K, CCO $72K, interns $30K). Health insurance $29K (4 W-2 × $600/mo). Compliance $162K. Infrastructure $54K. Operations $175K.
Year 2 Operations	45.4%	$1,361K — Salaries $750K (merit raises if performance warrants + intern ramp), health insurance $29K, compliance $240K, office $48K, cloud $72K, benchmark compute $36K, accounting $24K, travel/BD $42K, insurance $18K, misc $102K.
Strategic Reserve	12.6%	$379K — Fundraise bridge $150K, unexpected legal $100K, emergency infra $80K, board/advisor expenses $49K.

Monthly Burn & Runway

Category	Mo 1–3	Mo 7–9	Mo 10–12
Salaries (CEO + 3 Eng + CCO + interns)	$56.4K	$58.2K	$58.7K
Health Insurance (4 W-2 × $600/mo)	$2.4K	$2.4K	$2.4K
Compliance (CFTC, IOSCO, calc agent)	$14.5K	$17.5K	$19.0K
Infrastructure (cloud, office, compute)	$7.3K	$10.0K	$12.0K
Operations (accounting, travel, misc)	$4.0K	$7.7K	$8.7K
Monthly Total	**$84,550**	**$95,750**	**$100,750**

Key line items: Fractional CCO $6K/mo, health insurance $2.4K/mo (4 W-2), Vanta SOC 2 $1.7K/mo (from M4), E&O/D&O $1.25K/mo, calculation agent $8K/mo (from M7), private office $4,000/mo.

- **Target monthly burn:** Year 1 average operational burn: $92K/month (excluding one-time startup costs)
- **Runway at zero revenue:** 30 months at zero revenue
- **Cash-flow positive target:** Month 18–24 at base case

Even with zero revenue, $3M gives 30 months of runway. With base-case API revenue starting Month 4, runway extends past the 24-month planning horizon. This is not a company that runs out of money before proving the model.

Revenue Projections — Month 12

Stream	Count	MRR
Pro subscriptions × $3K/mo	8–12	$24–36K
Enterprise subscriptions × $10K/mo	2	$20K
Terminal licensing (partial)	—	$10K
Total exit MRR		**$54–66K**

Conservative estimate. Derivatives settlement revenue is excluded entirely — treated as pure upside. OTC licensing (minimum $100K/yr per licensee) is not included in the 12-month projection. First OTC swap target: Month 20–24.

Cap Table — Pre-Seed and Post-Seed

Holder	Pre-Round	Post-Seed
Founder (Bear Januszewski)	90%	72%
Seed Investors	—	20%
Employee Option Pool	10%	8%

Option pool: 10% (1M of 10M authorized shares), 8% post-dilution. Allocated to 3 engineering hires (2.0% + 2.0% + 1.5%, 4yr vest, 1yr cliff), fractional CCO (0.5%, 2yr vest), and 2.0% reserved for future hires/advisors. No co-founders — clean cap table.

Path to Series A

Parameter	Target
Trigger	OTC swap volume + cash flow positive + Bloomberg tickers live
Timeline	Month 20–24
Raise	$15–25M at $80–120M valuation
What you'll have	Bloomberg/Refinitiv tickers live, IOSCO governance framework, 2–3 OTC bank licensees, all datacenter-grade accelerators benchmarked, 12+ months degradation data, patent prosecution complete

THE TEAM

Founder

Bear Januszewski. Quantitative finance background. Dallas, TX.

Solo-built the entire platform in under 6 months: patent-pending normalization methodology, benchmark suite (bm-suite), data collection pipeline across 28 cloud providers, real-time API with 34 endpoints, TimescaleDB with 5.6M+ price ticks, and MVP dashboard. Goldman Sachs Futures Sales Trading discussions (Managing Director + VP level) — two meetings completed.

Yggdrasil Financial Technologies, Inc. (Delaware C-Corp, filed independently). Wyoming LLC retained as holding entity.

Hiring Plan (Post-Funding)

Role	Timeline	Focus
Benchmark/Infra Engineer	ASAP	Systematic GPU benchmarking across providers. Degradation curve infrastructure. eCU pipeline. Docker automation for spin-up → benchmark → teardown.
Backend/API Engineer	ASAP	Enterprise API hardening. WebSocket real-time feed. Bloomberg/Refinitiv integration. New provider collectors. SOC 2 technical controls.
Frontend Engineer	ASAP	Dashboard as "Bloomberg for compute." Customer portal. Data visualization. eCU comparison views. Degradation curve display.
Fractional CCO	ASAP	Manages all 6 compliance workstreams — CFTC opinion, IOSCO framework, Bloomberg distribution, calc agent oversight, E&O/D&O renewals, SOC 2. 15–20 hrs/wk.

Compensation: Backend/Infra engineers at $150K base + 2.0% equity each. Frontend engineer at $140K base + 1.5% equity (4yr vest, 1yr cliff). Fractional CCO at $72K/yr ($6K/mo, 15–20 hrs/wk) + 0.5% equity (2yr vest). Remaining 2.0% reserved for future hires/advisors. Competitive

equity compensates for below-peak-market cash in Dallas cost of living. At $100M Series A valuation, 2.0% equity = $2.0M (before dilution). At CME listing, life-changing.

Recruiting channels: UT Dallas, SMU, Texas A&M CS programs. Stanford CS recruiting pipeline. HPC labs. Ex-NVIDIA/AMD systems engineers. GPU/ML engineering communities.

COMPLIANCE ROADMAP

Yggdrasil is an index provider, not a swap dealer or commodity pool operator. No CFTC registration is required. The compliance burden is documentation, governance, and legal opinions — not regulatory licensing. A fractional Chief Compliance Officer manages all six workstreams from Day 1.

#	Workstream	Timeline	Budget	Outcome
1	CFTC Commodity Opinion	Month 1–3	$35K	Written opinion: CU = exempt commodity under CEA §1a(20)
2	E&O + D&O Insurance	Month 2–3	$15K/yr	$5M per-occurrence E&O, $5M D&O coverage
3	IOSCO Governance Framework	Month 3–6	$50K	19-principle self-assessment, methodology whitepaper, oversight committee (4 members)
4	Vanta SOC 2	Month 3–9	$15K setup + $20K/yr	Type I certification by Month 9
5	Bloomberg Distribution	Month 3–12	$15K setup + $3K/mo	Tickers YGCU, YGTCU, YGICU live on Bloomberg terminal
6	Calculation Agent	Month 6–9	$7.5K setup + $8K/mo	Independent daily re-verification (MerQube or boutique quant)

ISDA Documentation (Month 9--18)

Once the CFTC opinion and calculation agent are in place, Yggdrasil's counsel drafts a bespoke ISDA confirmation template. The first OTC counterparty's bank (target: Goldman Sachs) drafts the confirmation; Yggdrasil reviews. Budget: $10K legal review. This unlocks the OTC licensing revenue stream.

COMPLIANCE COST SUMMARY

$158K one-time startup costs — $65K compliance (CFTC opinion, E&O/D&O, SOC 2 setup), $93K infrastructure (office buildout, legal/IP, on-prem backup server, workstations, hiring, $25K buffer). Compliance recurring costs begin Month 7 at $23.5K/month, funded from Year 1 operations. Year 2 ongoing: $23.5K/month. Total 24-month compliance spend: $530K (18% of raise). Managed by fractional CCO ($72K/yr, 0.5% equity) — not Bearj's time.

Why This Matters

Every compliance milestone unlocks the next revenue tier. The CFTC opinion unlocks Bloomberg conversations. Bloomberg distribution unlocks Enterprise customers. The calculation agent unlocks settlement-grade data. ISDA documentation unlocks OTC licensing — the $250K–750K/licensee revenue stream that drives the Series A thesis.

THE ASK: $3M SEED

$3M post-money SAFE at $15M cap. 20% dilution. Clean, founder-friendly structure — no board seats, no liquidation preferences, no priced-round complexity.

Milestones This Unlocks

Milestone	Timeline	Phase
3 engineers + fractional CCO, office operational	Month 1–3	Infrastructure
15+ providers collecting reliably	Month 3	Infrastructure
Enterprise API live with rate limiting + auth	Month 3	Infrastructure
First paying customer	Month 4	First Revenue
Bloomberg/Refinitiv submission	Month 9–15	Credibility
IOSCO governance framework in place	Month 9	Credibility
First degradation curves published	Month 6+	Differentiation
Cash-flow positive (base case)	Month 18–24	Sustainability
First OTC swap referencing CU	Month 20–24	Derivatives
Series A raise	Month 20–24	Scale

Why $3M / Why Now

- **$45M+ raised by competitors** validates the market — VCs are funding this space actively
- **Goldman conversations occurred at MD/VP level** — funding enables follow-up. Capital commitment signals institutional viability. Reactivation not guaranteed.
- **Patent priority secured** — need to execute before competitors add normalization methodology (SiliconMark is already benchmarking, one step away from normalization)
- **AWS 15% price hike (Jan 2026)** proved enterprises need hedging tools — demand is tangible and growing
- **$3M gives 30 months runway** at zero revenue — plenty of time to reach profitability on API subscriptions alone, without touching derivatives revenue

LONG-TERM VISION

Once CU is adopted as the settlement benchmark, it becomes self-reinforcing. Every contract, every derivative, every loan covenant that references CU increases switching costs exponentially. The VIX Index built CBOE into a $17.8 billion company.[18] S&P earns $500M+/yr from index licensing alone. Once you are the benchmark, you are the benchmark forever.

Year 3+ opportunity:

- **CME-listed compute futures** — per-contract royalty revenue at scale. The VIX playbook: index → futures → options → ETFs → structured products.
- **Sector sub-indices** — GPU Index, TPU Index, ASIC Index as separate tradeable benchmarks. Each becomes its own derivatives market.
- **Degradation data as premium product** — monthly re-benchmarks reveal GPU aging curves. Trading firms pay $25–50K/mo for performance decay data that informs futures curve construction, insurance pricing, and fleet valuation.
- **China expansion** — trillions in Asian compute spend with zero hedging infrastructure. Founder learning Mandarin with existing Shenzhen contacts.
- **Provider partnerships** — quarterly permanent benchmarking agreements with cloud providers. They get fleet quality data; Yggdrasil gets exclusive measurement access and sellable degradation curves.
- **Compute insurance products** — CU-referenced coverage for performance degradation, provider outages, and spot price spikes.

At maturity: $300–500M/yr in derivatives settlement revenue. This is the VIX model applied to a $2.5 trillion underlying market.[19] Compute derivatives represent an entirely new asset class — and Yggdrasil owns the standard that denominates it.

RISK ANALYSIS AND MITIGATIONS

Risk	Level	Mitigation
Competitor develops normalization	Medium	Patent priority date secured. Non-provisional filed. CIP planned for v2 calibrated exponent methodology. 12+ month head start with working product.
Goldman loses interest	Low	Dual-path: direct-to-market via Bloomberg/Refinitiv distribution. Goldman is upside, not dependency. API revenue stands alone. Funding enables follow-up; reactivation not guaranteed.
Slower market adoption	Medium	Profitable on data API alone by Month 12. Derivatives are pure upside. 30 months runway at zero revenue. Not fragile.
Benchmarks beat spec-based	Low	Complementary, not competitive (grade spec vs QA). Yggdrasil does BOTH — CU (spec-based) and eCU (benchmark-based). SD data feeds into CU calibration.
Regulatory barriers	Medium	OTC swaps are bilateral — no CFTC approval needed. Exchange listing is Phase 2. Cash-settled contracts have clearest regulatory path (Commodity Exchange Act § 1a(9)).
Patent claims rejected	Low–Med	Claims are Alice-resistant: tied to physical hardware measurement, not abstract math. Even without grant: 12+ month head start, working product, customer relationships create switching costs.
Solo founder risk	Medium	Solo-built entire platform in under 6 months. 3 engineers + fractional CCO from Day 1. Key-person risk decreases rapidly post-hire. Code is well-documented and modular.

GPU compute today = oil in 1982. CU is the standard.

Bear Januszewski · bearj@yggfin.tech
Yggdrasil Financial Technologies · Dallas, TX · Patent Pending

APPENDIX

A. Detailed Financial Projections

Revenue Projections

Detailed quarterly revenue build available in financial-model.md. Year 1 collected revenue: $120–300K (bear to base). SOM of $180K represents full addressable Year 1; actual collection depends on customer ramp timing. See Financial Model for month-by-month projections.

Runway Scenarios

Scenario	24-Mo Burn	Remaining Cash	Runway
Zero revenue	$2,621K	$379K	30 months
Conservative ($600K rev)	$2,621K	$979K	36+ months
Base case ($1.5M rev)	**$2,621K**	**$1,879K**	**Indefinite**
Upside (OTC + $3M rev)	$2,621K	Break-even+	Indefinite

Monthly Burn Detail

Line Item	Monthly	Assumption
Backend/Infra Engineer	$12,500	$150K/yr
Benchmark/API Engineer	$12,500	$150K/yr
Frontend Engineer	$11,700	$140K/yr
Founder (CEO)	$11,700	$140K/yr
Fractional CCO	$6,000	$72K/yr, 15–20 hrs/wk, 0.5% equity (2yr vest)
Health Insurance (4 W-2)	$2,400	$600/mo per W-2 employee (CEO + 3 Eng)
Interns (1–2, 1099)	$2–4K	Part-time, project-based ($30K/yr budget)
Compliance (amortized)	$14.5–19K	CFTC, IOSCO, E&O/D&O $1.25K, Vanta SOC 2 $1.7K, calc agent $8K (from M7)
Private Office	$4,000	Dedicated office, Dallas
Cloud + Benchmark Compute	$3–7.5K	Pipeline + API + GPU instance runs
Operations (acct, travel, misc)	$4–8.7K	Kruze, Gusto, BD, travel, tools
Total	**$85–101K**	**Average: $92K/month**

B. Patent Claims Summary

The patent application covers the following claim categories. Full claims are available under NDA in the data room.

#	Category	Scope
1	Core Method	Multi-parameter normalization from hardware specifications to produce deterministic accelerator scores
2	System	Hardware + software system for cross-architecture compute scoring
3–4	Workload Taxonomy	TCU (training) and ICU (inference) as separate workload-optimized scores with physics-constrained weighting
5	Immutable Reference	Fixed reference accelerator principle — scores are permanent per hardware generation
7–11	Specialized Indices	Workload-specific compute units for rendering, HPC, genomics, and other domains
13	Multi-Architecture	GPU (CUDA), GPU (ROCm), TPU (MXU/JAX), ASIC, NPU, FPGA coverage
15	Forward Compatibility	Methodology extensibility to future accelerator architectures without formula changes

Summary only. Exponent values, calibration methodology, and proprietary weight optimization techniques are trade secrets not disclosed in this document. Full patent text available in the data room under NDA.

C. Benchmark Methodology Overview

The **bm-suite** benchmark system measures five performance dimensions on physical hardware:

Dimension	What It Measures	Method
GEMM FLOPS	Raw compute throughput at FP32, FP16, BF16, FP8	Matrix multiply kernels via cuBLAS/hipBLAS
Memory Bandwidth	HBM read/write throughput	Stream-style bandwidth test
VRAM Capacity	Usable accelerator memory	Progressive allocation test
Interconnect	NVLink / ICI / fabric throughput	All-reduce bandwidth measurement
Inference	End-to-end token generation speed	vLLM serving on standard models (Llama 70B, 8B)

Key design principles:

- **Thermal gating:** benchmarks wait for thermal equilibrium before measurement, preventing cold-start inflation
- **Docker isolation:** each benchmark runs in a clean container — reproducible across providers
- **Parallel GPU runner:** benchmarks multiple GPUs simultaneously within a node
- **Calibration scripts:** cross-reference bm-suite results against MLPerf and vendor specifications

bm-suite produces **eCU scores** — empirical performance measurements that are compared against spec-derived CU to calculate the CU–eCU spread. This spread is the premium data product: provider-specific, time-varying, and commercially valuable for degradation curve construction.

Benchmark kernel implementations, scoring algorithms, and calibration coefficients are proprietary. The methodology is described at a level consistent with public patent disclosures. Detailed technical documentation is available in the data room under NDA.

SOURCES AND CITATIONS

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11. CBOE Global Markets, "VIX Index History: Launched January 19, 1993."

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15. Bloomberg, "Silicon Data Creates First-of-its-Kind Index for AI Chips," May 27, 2025. SDH100RT index launched at $2.37/hour.

16. Derived from 53.7M annual VIX contracts (CBOE, 2024) at $0.10/contract royalty. CBOE market cap: $17.8B. Burton-Taylor International Consulting estimates global index industry revenue at $5B annually (2024), with industry gross margins of 70–80% (MSCI annual reports, Macrotrends analysis).

17. Bloomberg, “Silicon Data Creates First-of-its-Kind Index for AI Chips,” May 27, 2025; Silicon Data, “Daily GPU Rental Index Launch,” Press Release (May 2025).

18. CBOE Global Markets market capitalization (2025). From 449 contracts on Day 1 (March 2004) to 53.7M contracts annually (2023). VIX futures, options, and related products generate the majority of CBOE’s revenue.

19. Gartner, “Global AI Spending Forecast” (2025). $1.5T in 2025, $2.5T in 2026. GPU compute underpins the majority of this spend as the core infrastructure commodity.